

# NUMIC — Accelerator application brief

## 1. Business value proposition and model

### Problem

In preterm babies, **bleeding in or around the brain** sometimes occurs (**IVH**). **When the brain's fluid-filled spaces enlarge beyond what teams expect (post-haemorrhagic ventricular dilation, PHVD)**, care depends on **how large those spaces are, how fast they change on repeat head ultrasounds, and how the baby is doing overall**. Teams **already build a risk picture** from the latest scan, earlier scans, and the bedside story—but **most of that work is done manually**: gathering the right measurements, **comparing them with the prior scan**, and **weighing that trend with the clinical picture** take **time and sustained attention** on a busy neonatal unit.

### Proposition

NUMIC is **decision support software**. It turns the **same inputs staff already use**—standard head-ultrasound numbers (how wide key fluid spaces are), **change since the last scan**, plus a **short, structured note on clinical concern**—into a single **risk band** (low / moderate / high), using transparent rules we call **NumicFlow**. **The score and band are explainable**: the output shows **how each part contributed** (current size, change since the prior scan, clinical modifier) and **which rule version** was applied, so the reasoning stays **visible and repeatable**, not a black box.

NUMIC **does not choose treatment**. Staff use the band to **match local pathways** (how often to scan, when to escalate, when to involve neurosurgery, and so on). **The team and the hospital stay in charge of decisions**.

### Value

- A **shared, written risk level** that is **explainable**—**plain reasons** and a **per-part breakdown**, so everyone means the same thing at handover.
- A **clear record** of what drove the band (breakdown + rule version) for governance and learning.
- A route to **fit into hospital workflows** over time (starting with measurements staff already record; any **smarter imaging step** would come later, with proper validation).

**Business model**: B2B SaaS—**annual licence per NHS trust or neonatal network**, with **optional pilot pricing** in year one and **separate setup / integration fees** where needed.

### Why this matters (clinical context—not a product proof)

Experts link **PHVD** to **patterns of brain injury** and **weaker long-term development** in preterm infants. In practice, care leans heavily on **repeat head ultrasound**—both **the size of the fluid spaces** and **whether they are getting worse**—so teams can act **in good time** and along **consistent**

local pathways.

Separately, **large national studies** show that children born **moderately or late preterm**—who make up **most preterm births**—can still face **higher long-term neurodevelopmental risk** than full-term peers. That is **background** on why neonatal brain care is a **public health scale** issue; it does **not** by itself prove anything about PHVD tools. **Supporting publications** at the end of this brief give optional reading.

What NUMIC adds is **less repetitive manual work**: staff do not have to **trace back through prior scans and dates to re-check every measurement** against **agreed cut-offs**—for example **high-percentile lines (such as the 97th)** or **fixed millimetre limits**—before a consistent risk picture is clear. **The unit’s agreed protocol is written into versioned rules**; NumicFlow **applies that bundle** and returns an **explainable risk band** (sub-scores and rule version visible)—**always** with normal **clinical judgment** still in charge.

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## 2. Technology — plain-English overview

NUMIC is **software** that takes **trusted ultrasound measurements** from head scans, **when each scan was done**, and a **short ward-side note** on how worrying the overall picture is. It runs **clear, written rules** (“NumicFlow”) to produce a **risk band** (low / moderate / high). **That is the same judgment neonatologists already make** from “then vs now” on the scans and the baby’s course; NUMIC puts it **on the page, consistently**, labels **which version of the rules** was used, and shows **what drove the score**—so the result is **explainable**, not opaque.

**The software does not prescribe treatment**—it supports **local escalation and follow-up practice**.

The design allows data to come in **manually today** and from **hospital systems later**. A **browser demo** shows how a band is produced; the **real product** is a proper clinical deployment with governance—not the demo on its own.

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## 3. Technology — extra detail (optional for reviewers)

*Skip this section if you only need the business story; it is here for due diligence.*

- **Prototype build:** A small web service (Python / FastAPI) that accepts measurement records and returns **NumicFlow** scores and bands; an HTML/JS demo runs alongside for the same system.
- **NumicFlow (logic):** One total score (**0–14**) from (i) **how big** key fluid-space measures are on the current scan (**0–6**), (ii) **how much they**

have moved since a **previous** scan (**0–6**), (iii) a **clinical concern** slot (**0–2**). The **risk band** (low / moderate / high) comes from **versioned** cut-points on that total. **Outputs stay explainable:** each layer’s points are surfaced with the total, not hidden. **No treatment text** is emitted—only the band and breakdown.

- **Rule versions:** Rules are packaged under **named versions** (e.g. `numic_flow_v1`) so results stay tied to a **specific rule set** as guidance updates.
- **Inputs today:** Typed entry, spreadsheet-style import, or simple feeds of numbers; **automatic “draw on the image and measure”** is **future work** and needs validation before clinical use.
- **Roadmap:** Proper clinician UI, data storage, **hospital imaging/workflow** links where needed; any **AI on images** only after **clinical validation**.
- **Regulatory / governance (open, not final):** Device **classification and market routes** (for example UK and EU frameworks and whether NUMIC is **software as a medical device**, and in which class) are **still to be settled** with clinical and quality advisors. The same conversations need to pin down **intended use** (who it is for, what it may and may not claim), a **clinical validation** approach before use near patients, **risk management** (e.g. wrong inputs, wrong rule version, misunderstood band), **software lifecycle and cybersecurity** for a connected product, **data protection** if identifiable patient data are processed, and **usability** so teams read outputs consistently on busy wards.

*No confidential model weights, unpublished trial data, or undisclosed third-party IP beyond what is described above.*

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## 4. Brief summary of key team members

| Name               | Role / expertise     | Relevant background                                                                                                                                                   |
|--------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eswari Mathialagan | Software development | Software developer; Master’s student at UCL;<br><a href="https://www.linkedin.com/in/sankareswarymathialagan">https://www.linkedin.com/in/sankareswarymathialagan</a> |

## 5. Investments and grants raised to date

None to date

## 6. Investment sources (non-dilutive, angel/syndicate, VC, corporate venturer)

None to date

## 7. Pre- or post-revenue

**Pre-revenue** — no commercial or pilot revenue recorded to date.

## 8. Present location of company (or virtual)

**Virtual** — no fixed operating premises; team and work are location-independent.

## 9. Interest in Discovery Park

Primary interest is the **pathway to NHS and clinical advisers**—including design-partner conversations, neonatology input, and regional health-innovation networks that sit naturally around Discovery Park and the wider Kent / South East cluster. NUMIC is **neonatal decision support**; credible iteration depends on **trusted clinical voices** as much as engineering.

## 10. Benefits of Discovery Spark

NUMIC's next stretch is **part clinical, part regulatory**—not only engineering. The benefits we are looking for from Spark are **practical rather than abstract**:

- **Regulatory introductions:** Access to people who understand **UK/EU health-software regulation, software as a medical device**, and **proportionate** routes from prototype toward **responsible clinical use**—so **intended use**, classification, and evidence planning are informed early, not improvised alone.
- **Clinical advisors (and design-partner conversations):** Introductions to **neonatology, imaging, and NHS governance** perspectives who can **stress-test** NumicFlow against real **ward workflow, handover, and local pathways**—before wider pilots or stronger claims.

Broader programme benefits (mentorship, ecosystem, credibility, facilities) still matter; the two bullets above are the **main gaps** we hope Spark helps fill.

## Elevator line

*NUMIC helps neonatal teams turn **repeat head ultrasound measurements** and bedside context into a **clear, explainable risk band** when **IVH is followed by PHVD**—so **next steps follow local protocol**, with every **driver of the score** visible, not hidden. It supports teamwork and record-keeping; it **does not** replace clinical judgment.*

### Supporting publications (optional background)

Mitha A, et al. Neurological development in children born moderately or late preterm: national cohort study. *BMJ*. 2024;384:e075630. <https://doi.org/10.1136/bmj-2023-075630>

El-Dib M, et al. Management of Post-hemorrhagic Ventricular Dilatation in the Infant Born Preterm. *J Pediatr*. 2020;226:16-27.e3. <https://doi.org/10.1016/j.jpeds.2020.07.079>

*This brief does not constitute regulatory advice or a performance commitment. Align final submissions with your legal and clinical advisors.*